

Theory que-ans

submitted by : faizkhan

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Practical & theory Exam

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Submitted By:- Faiz Khan Pathan

Part: A :- Theory (Short ques)

Q-1 Define Mean, Median, Mode with an Example in context of Employee Salaries

Ans → mean, median, mode :- in the context of Salaries, mean is the mathematical average.

→ median is the exact middle salary when sorted.

→ mode is the most frequently occurring salary.

Q-2 Diff b/w Range and variance.

→ Range is simply highest salary minus lowest salary. variance measure how spread out "all" salaries are from the average high variance means extreme inequality.

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Q3 Diff b/w normal Distribution & Poisson Distribution

→ Normal Diss represents continuous, symmetric data like employee performance score.

→ Poisson Diss represents discrete count of events like the number of projects completed.

Q4 Explain skewness with one workplace related example.

Ans → Skewness measures asymmetry, workplace example: Corporate salaries are usually right-skewed because most staff earn a base wage while a few executives earn millions.

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Q5 Define conditional Probability and explain how it applies in Promotions.

Ans. Conditional Probability: This is the chance of an event occurring given another event has already happened. In Promotions it helps us answer "What is the Probability of Promotion given the employee's performance score is strictly > 80 ?"

Q-6 Explain independent vs mutually Exclusive Events with Example.

Ans. Independent: Events that don't affect each other.

Mutually Exclusive: Events that can't happen simultaneously.

Q7. Why Does Bayes theorem helps us in the real-world decision making.

Ans → Bayes theorem: Helps us update probabilities based on new evidence. E.g. - If we know 5% of employees quit but an employee suddenly takes 5 sick days, Bayes helps calculate the updated probability then about resigning.

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Q8 Write ~~short~~ short note on (PCA) in
Simple words

Ans → PCA: A technique to simplify
the large dataset. if we have
so overlapping performance metrics,
PCA compress them into a few
non overlapping principal components.
about capturing the core information

Practical output screenshot.

Step 1: Central Tendency & Dispersion

Salary - Mean: \$82865.68, Median: \$73081.53, Mode: \$9633.24
Projects Completed - Variance: 4.92, Standard Deviation: 2.22

Step 2: Probability & Events

General Probability of promotion: 36.73%

Contingency Table (Promotion vs Dept):

Department	Finance	HR	IT	Marketing	Sales
Promotion_Status					
No	485	500	519	515	512
Yes	332	268	310	294	265

Conditional Probability -> $P(\text{Promotion} \mid \text{Performance} > 80)$: 87.48%

Step 3: Distributions & Visualization (Charts)



Step 4: Linear Algebra Application

Taking Emp 0 Vector = [3 40.7], Emp 1 Vector = [7 40.5]
Dot Product between Emp 0 & Emp 1: 1669.35
Emp 0 -> Norm 1 (Manhattan): 43.70
Emp 0 -> Norm 2 (Euclidean): 40.81